



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.5

Powers and Exponents

Flagship

Lesson 6-1-flagship · Teacher Copy

Name: _____

Date: _____

Class: _____

SOUND STUDIO MISSION

The Doubling Mixboard

You are the lead sound engineer at Amplitude Studios. Every time you flip a switch on the new mixboard, the volume doubles — that is the power of exponents. The headline act goes on in one hour, and you must master powers before the speakers can be safely dialed in.



TODAY'S OBJECTIVES

Content Objective: I can write and evaluate numbers in exponent form using a base and a power.

Language Objective: I can explain my work using the words exponent, base, power, and evaluate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Exponent

Base

Power

Evaluate



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The small raised number that shows how many times the base is multiplied is the

2

The number being multiplied repeatedly in a power, like the 3 in 3^4 , is the

3

A number written with a base and an exponent, like 3^4 , is a .

4

To find the value of an expression, like $2^3 = 8$, is to it.

Reflect — Exit Ticket

What is the value of 3^4 ?

- A. 81
- B. 12
- C. 34
- D. 64

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Exponent
2. **Notes 2:** Base
3. **Notes 3:** Power
4. **Notes 4:** Evaluate
5. **Watch for this mistake:** *A common mistake in Powers and Exponents is skipping the key idea: "A power like 2^3 is a short way to write repeated multiplication: $2 \times 2 \times 2$." — always check your work against this rule before you submit.*
6. **Try It 1:** A. 25 — 5^2 means $5 \times 5 = 25$, so 25 unlocks Track 2.
7. **Try It 2:** A. 10 — *The base is the number being multiplied repeatedly. In 10^4 the base is 10 and the exponent is 4, so $10^4 = 10 \times 10 \times 10 \times 10 = 10000$.*
8. **Exit Ticket:** A. 81 — $3^4 = 3 \times 3 \times 3 \times 3 = 81$.